

2(a)	too large / large size ; A not small enough cannot pass through endothelial, pores / gaps I holes A fenestrations / AW or cannot, pass across / cross the membranes of, endothelial cells ; A epithelial cells <i>in context of entering an endothelial cell and exiting to the tissue fluid</i>	2
2(b)	biuret (solution / reagent) ; A copper sulfate solution and, sodium / potassium, hydroxide (change from light) blue to, lilac / purple / violet / mauve ;	2

Question	Answer	Marks
2(c)	<i>accept Hb / hb for haemoglobin</i> any three from: (at altitude) lower, partial pressure of oxygen / pO_2 (in, atmosphere / inhaled air / inspired air / alveolar air) ; I low pO_2 in blood R if stated as high $ppCO_2$ and low $pp O_2$ R low, volume / saturation, of oxygen percentage saturation of haemoglobin (with oxygen), lower / decreased ; A haemoglobin is less saturated A fewer molecules of / less, oxygen, combine / associate, with haemoglobin I absorbed / taken up by R if in context of Bohr effect but then allow ecf for overall max 2 (as) haemoglobin has lower affinity for oxygen (than at sea level) ; more haemoglobin (synthesised / required / provided) ; compensation / described ; e.g. helps to transport the same quantity of oxygen as at sea level AVP ; e.g. <i>idea of</i> more rbc through pulmonary capillaries per unit time <i>ref. to</i> EPO / erythropoietin (secreted to increase red blood cell production)	3

Question	Answer	Marks
2(d)	any three from: A Ψ for water potential A plasma for blood <i>mp1 and mp2 = consequence in blood of lower blood albumin than normal</i> 1 <i>idea that</i> low blood albumin means, less / AW, solute ; 2 (so) higher water potential (in blood); <i>must be in context of blood context can be anywhere along the capillary</i> A high water potential <i>if ref. to low albumin is made</i> A steeper water potential gradient (than normally, at arterial end) <i>mp3 and mp4 = return of water to blood from tissue fluid (capillary venous end)</i> 3 less steep water potential gradient (than normal) / little difference in water potentials ; 4 less / little, water returns to, blood / capillary (from tissue fluid) ; A (tissue) fluid <i>for water</i> R plasma <i>for water</i> <i>additional ideas</i> 5 <i>idea that</i> more, water / plasma / fluid, enters tissue fluid (than normal) R tissue fluid <i>for fluid</i> A more, water / fluid, enters tissue fluid than exits (to the capillary or lymph system) ; (<i>compared to normal</i>) 6 AVP ; e.g. water can enter tissue fluid for a longer time than normal less albumin than normal to act as an osmotic force for return of water / AW too much excess, water / fluid / tissue fluid, to be taken up into lymph, capillaries / vessels / system	3
2(e)	(cell surface) receptor(s) ;	1